Chapter 2

Space, Time, and the Eye of the Beholder

In June 1905, twenty-six-year-old Albert Einstein submitted a technical article to the Annals of Physics in which he came to grips with a paradox about light that had first troubled him as a teenager, some ten years earlier. Upon turning the final page of Einstein's manuscript, the editor of the journal, Max Planck, realized that the accepted scientific order had been overthrown. Without hoopla or fanfare, a patent clerk from Bern, Switzerland, had completely overturned the traditional notions of space and time and replaced them with a new conception whose properties fly in the face of everything we are familiar with from common experience.

The paradox that had troubled Einstein for a decade was this. In the mid-1800s, after a close study of the experimental work of the English physicist Michael Faraday, the Scottish physicist James Clerk Maxwell succeeded in uniting electricity and magnetism in the framework of the electromagnetic field. If you've ever been on a mountaintop just before a severe thunderstorm or stood close to a Van de Graaf generator, you have a visceral sense of what an electromagnetic field is, because you've felt it. In case you haven't, it is somewhat like a tide of electric and magnetic lines of force that permeate a region of space through which they pass. When you sprinkle iron filings near a magnet, for example, the orderly pattern they form traces out some of the invisible lines of magnetic force. When you take off a wool sweater on an especially dry day and hear a crackling sound and perhaps feel a momentary shock or two, you are witnessing evidence of electric lines of force generated by electric charges swept up by the fibers in your sweater. Beyond uniting these and all other electric and magnetic phenomena in one mathematical framework, Maxwell's theory showed—quite unexpectedly—that electromagnetic disturbances travel at a fixed and never-changing speed, a speed that turns out to equal that of light. From this, Maxwell realized that visible light itself is nothing but a particular kind of electromagnetic wave, one that is now understood to interact with chemicals in the retina, giving rise to the sensation of sight. Moreover (and this is crucial), Maxwell's theory also showed